

Flood Susceptibility Model — Defense Guide

A plain-language companion to [Flood_Susceptibility_Model_Report.pdf](#). Read this first to get the whole model in your head quickly; go to the full report only for the literature tables and appendix.

1. The 60-second summary

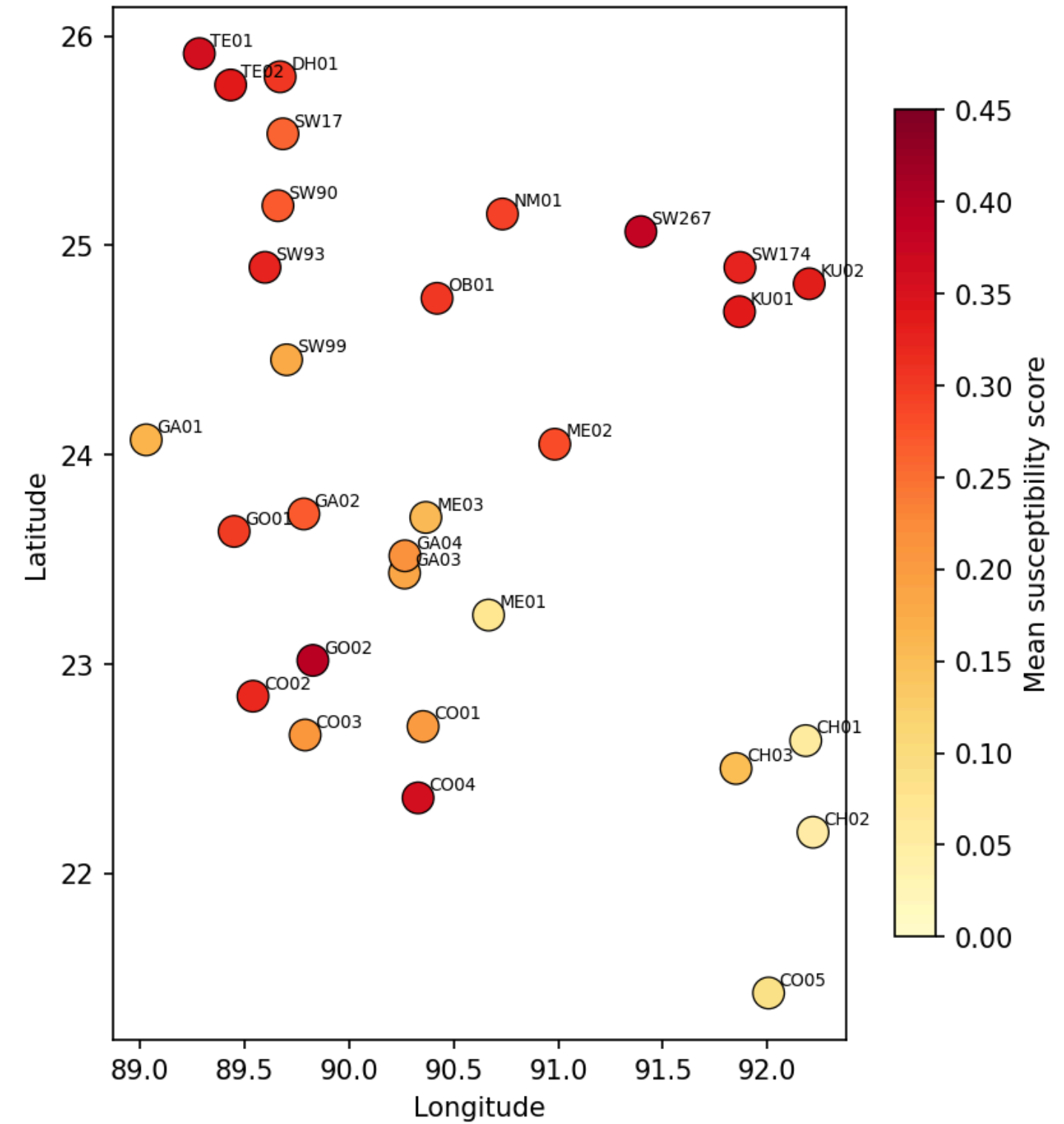
Models #1 and #2 both ask a moving question: *will conditions here get dangerous soon*, using live weather. This third model asks a **fixed** question instead: **is this specific patch of ground, by its own geography, the kind that floods — or the kind that stays dry** — independent of any particular storm, using only elevation, slope, distance to the nearest river, drainage density, and land cover. It doesn't compete with the other two models; it answers something they structurally cannot, since terrain doesn't change day to day. All three combine into one transparent formula (§6 below).

Final, honestly-evaluated result on stations the model **never saw during training**: **ROC-AUC 0.908, PR-AUC 0.785**.

2. How it works, in plain terms

1. **A local grid, not a national map**: a 7×7 grid (49 points, ~10km span) around each of the 30 monitored stations — 1,470 points total. Scoped to what this project actually monitors, not an attempt at a full Bangladesh atlas.
2. **Labels come from real satellite observation, not guesswork**: every point was checked against the **entire 2016–2026 Copernicus Sentinel-1 SAR archive** — 918,423 total real observations. A point is "flooded" if the archive ever caught it flooded once; "non-flooded" only if it was cleanly observed 200+ times with zero detections — a deliberately strict, *inspected* bar (median depth was 575 observations/point, comfortably above 200), not an arbitrary number.
3. **Terrain features, mostly self-derived**: elevation and slope come from Copernicus DEM GLO-30 via real flow-routing (pysheds); land cover from ESA WorldCover. Distance-to-river was self-derived originally, then **replaced after a real cross-check** (see Q4 — a good story for a defense).
4. **Model**: Random Forest — chosen after a real head-to-head against LightGBM, not assumed just because the other two models use LightGBM.
5. **No live computation at request time**: since terrain doesn't change day to day, every point's score was computed once at training time and saved to a lookup table — the live API is a fast table lookup, not a 30-60 second DEM-processing call.

30 monitored stations, colored by mean susceptibility score
(plain lat/lon scatter, not a GIS basemap)



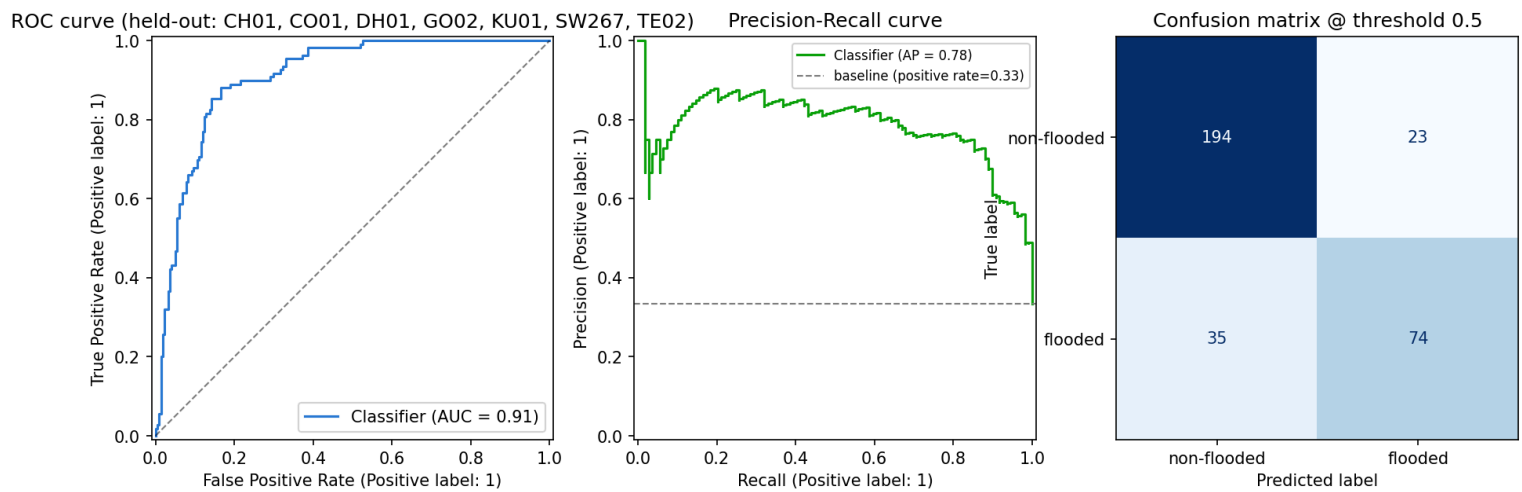
All 30 stations by real coordinates, colored by mean susceptibility — Meghna-basin haor/wetland stations score highest, Chittagong Hill Tracts stations lowest, matching known Bangladesh flood geography.

3. The numbers you'll be asked about

Evaluation	ROC-AUC	PR-AUC
Spatial GroupKFold CV, Random Forest	0.888 (mean)	—
Spatial GroupKFold CV, LightGBM	0.862 (mean)	—
Held-out test (7 stations, never touched in training)	0.908	0.785

7 of 30 stations — stratified across all 4 basins, so the held-out set still spans flat floodplain, haor wetlands, *and* hills — were held out completely, never touched during training or model selection.





ROC curve, Precision-Recall curve, and confusion matrix on the held-out spatial test set.

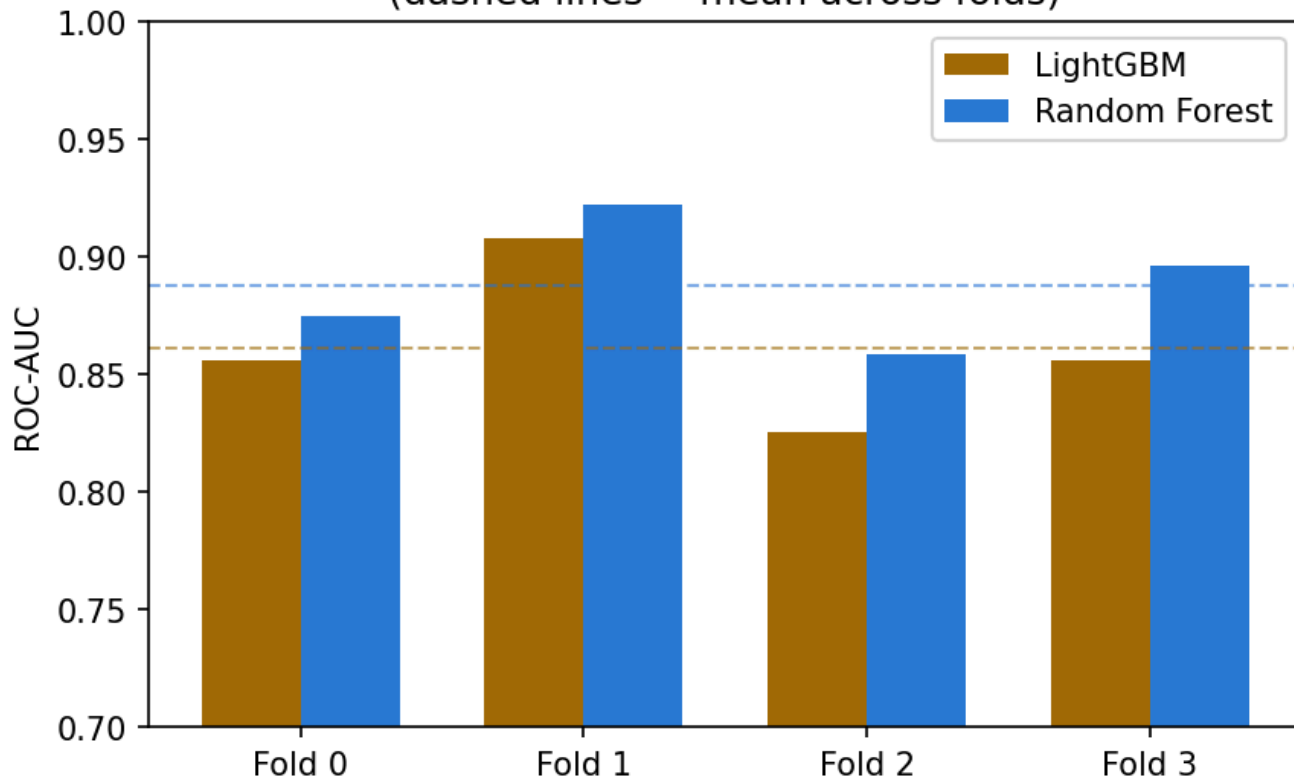
4. Likely defense questions, answered

Q1: Why does this model score "only" 0.91 ROC-AUC when other Bangladesh studies report 97%+ accuracy? Two separate reasons, both real, not excuses. First, accuracy and ROC-AUC measure different things — accuracy can look high on an imbalanced dataset even from a mediocre model. Second, and more important: random train/test splits on spatial data are **documented in the literature to inflate reported AUC by 5–15%**, because nearby points are correlated (if point A floods, its neighbor 200m away probably does too). This model's 0.908 comes from **7 whole stations held out completely** — the harder, more honest evaluation standard. Most reviewed studies don't disclose whether they controlled for this at all.

Q2: What is "spatial cross-validation" and why does it matter more here than a normal split would? A normal random split can put point A in training and its next-door neighbor point B in test — B is then trivially easy to predict because it's basically the same ground as something the model already saw, which inflates the score without proving real generalization. Spatial GroupKFold groups all of a station's points together in the same fold, so no fold ever trains on a point spatially close to one it's tested on. This project went further and additionally held out entire stations end-to-end.

Q3: Why Random Forest instead of LightGBM, when your other model uses LightGBM? Not assumed — tested head-to-head on identical data via spatial cross-validation (0.888 vs. 0.862 mean ROC-AUC, RF wins). This matches an independent, Bangladesh-specific literature finding that plain RF can beat gradient boosting specifically in hilly terrain — and this grid genuinely spans both the flat delta *and* the hilly Chittagong Hill Tracts, so the distinction was worth checking rather than assuming the first model's choice would transfer.

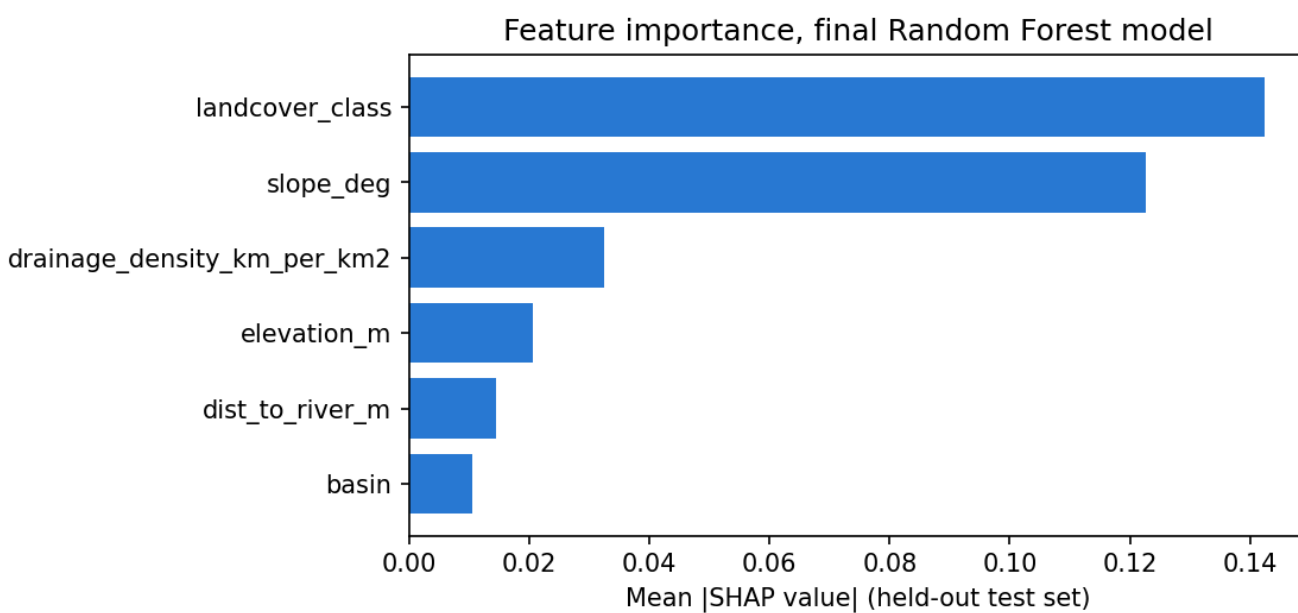
Spatial GroupKFold CV: LightGBM vs Random Forest
(dashed lines = mean across folds)



Spatial GroupKFold cross-validation: Random Forest wins on every fold.

Q4: Tell me about a mistake you caught and fixed — walk me through it. Two real examples, good material for a defense since they show real verification discipline, not just a clean narrative written after the fact:

- *Near-duplicate work avoided:* model #1 already has a peer-reviewed elevation/HAND terrain feature (from MERIT Hydro). Early in this model's design, before checking, a self-computed replacement was nearly built from scratch — which would have both duplicated existing work and produced a strictly *less* rigorous version of an already-correct feature. Caught before implementation.
- *A feature was cross-checked and swapped after the fact:* the self-derived distance-to-river feature (from DEM flow-routing) was compared point-by-point against HydroRIVERS, an independent, peer-reviewed global river network. The two barely correlated across the full grid (Pearson $r \approx 0.03$). Rather than assume either was right, **both were empirically tested through the full training pipeline** — identical spatial CV, identical held-out test, changing only this one feature. HydroRIVERS won by a modest but real margin on the held-out test (0.908 vs. 0.902 ROC-AUC) and rests on an externally validated dataset rather than a self-computed one, so production was switched to it. The original self-derived version is kept in the data for provenance, not silently deleted.



SHAP feature importance — land cover and slope dominate, consistent with flat deltaic terrain where these carry more signal than raw elevation.

Q5: How does this combine with the other two models? One transparent formula, not a black-box stacked meta-model:

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combined_risk = classifier_24h_probability × (0.5 + 0.5 × susceptibility_score)
```

Susceptibility acts as a **bounded modulator** ($0.5\times\text{--}1.0\times$) on the classifier's time-bound probability — it can reduce a forecast for inherently low-risk ground, but never fully zeroes it out, since an extreme enough storm can still flood low-susceptibility terrain. Chosen deliberately over a stacked model so the combination logic stays defensible in one line, not requiring its own separate justification.

Q6: Why trust a 200-observation cutoff for a "non-flooded" label? It wasn't picked in advance — it was chosen *after* inspecting the real observation-count distribution per point (median 575 observations, well above 200), specifically so the bar excludes very little real data while still requiring substantial evidence before trusting a negative. It's disclosed as a real limitation regardless (§10 of the full report): a small number of points near that exact boundary could be relabeled under a different, similarly defensible cutoff.

5. Weaknesses to own before they're asked

- Scoped to 30 station neighborhoods (1,470 points), not a national susceptibility atlas — deliberate scope, not a shortcut.
- Land cover is a single 2021 snapshot (checked directly: no newer ESA WorldCover release exists yet) — land-use change over time isn't captured.
- The 200-observation threshold is inspected and reasoned, but still a threshold (Q6).
- Drainage density is still self-derived from a single DEM via flow-routing — not independently validated against ground survey, though the underlying DEM itself is independently accuracy-assessed as one of the better free global options. Distance-to-river no longer carries this caveat (Q4).
- HydroRIVERS itself is DEM-derived (from a coarser $\sim 500\text{m}$ source), not ground-surveyed — an improvement in independence over the original feature, not an absolute ground-truth reference.
- The 7-station held-out test (326 points) is real but small — a larger held-out set would tighten the confidence interval around 0.908.

6. If they only remember one thing

A genuinely different, complementary question from the other two models — evaluated the hard way (whole stations held

out, never a random split), with two real examples of a design decision being checked empirically instead of assumed, and a combination formula simple enough to defend in one line.